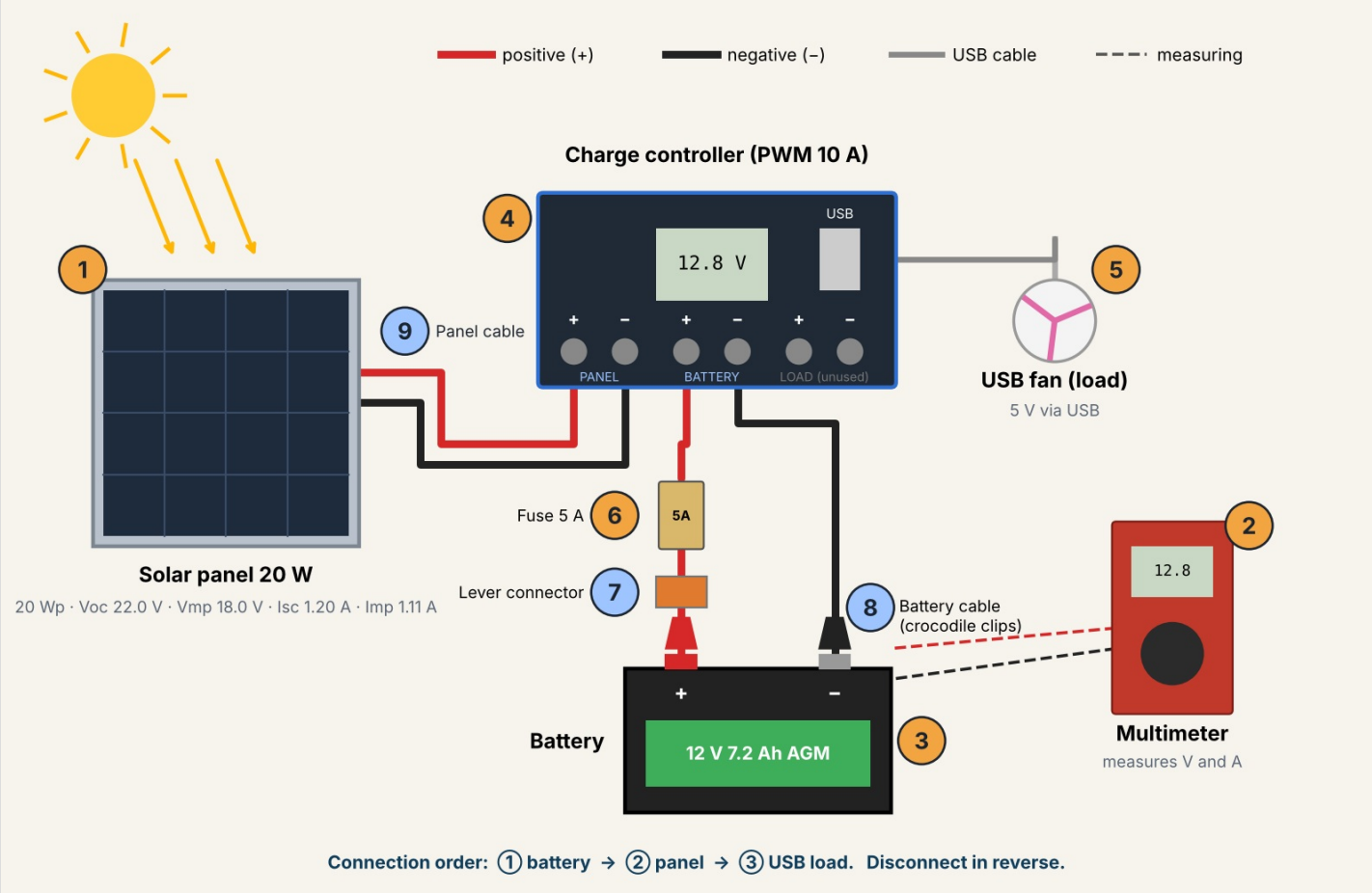
Solar + Battery Test System — Readings Log

Mohamed Eniw · Passau, Germany · October 2026 · values measured with an AstroAI AM33D multimeter

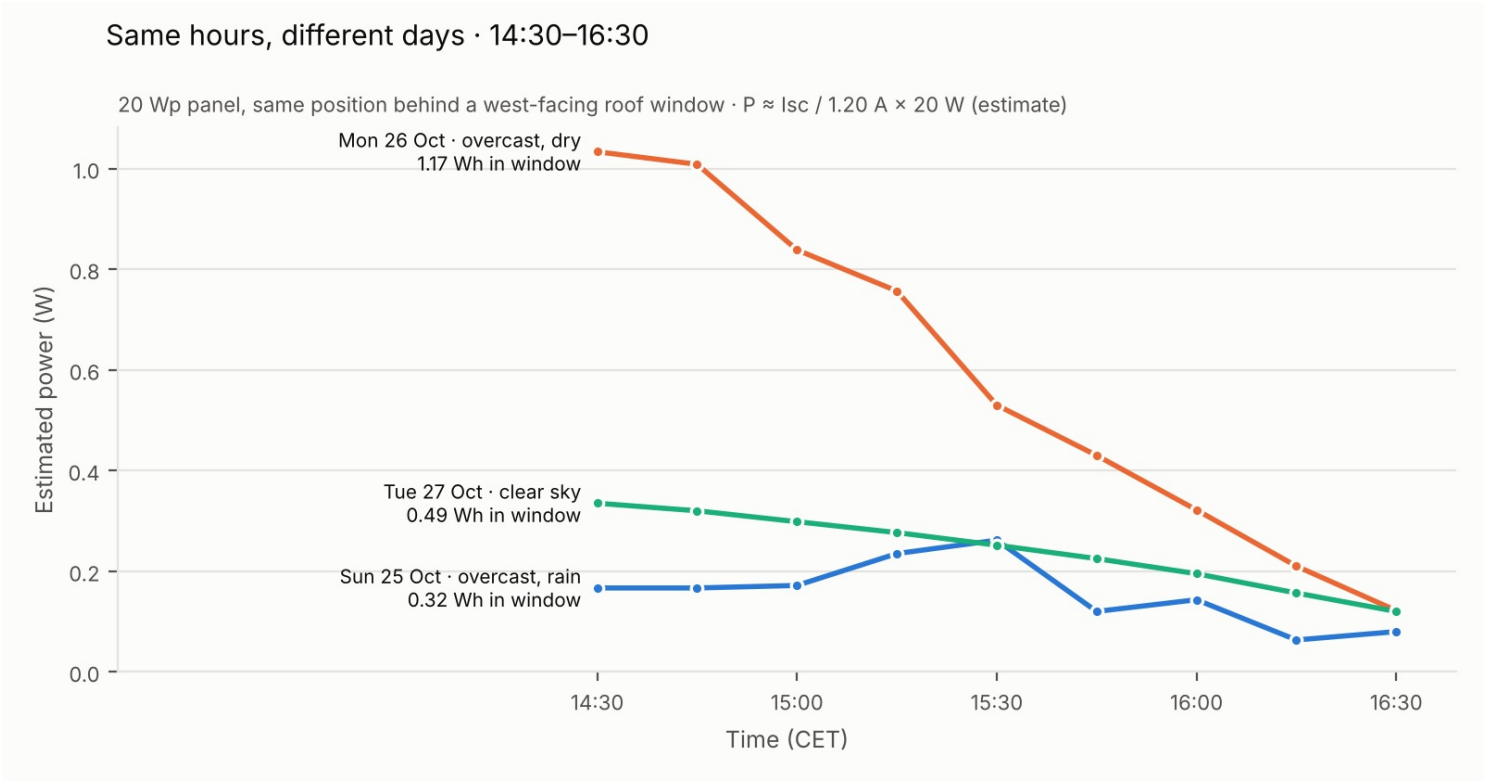
#	Measurement	Parts	Meter setting	Expected	Measured	Time / conditions
R1	Battery voltage at rest (before wiring)	2 on 3	V⎓ 20	12.6–13.0 V	12.94 V	Sat 24 Oct, 18:01, indoors
R2	Battery voltage shown on controller	4	display	≈ R1	12.7 V	Sat 24 Oct, 19:22, indoors
R3	Battery voltage with fan running	2 on 3	V⎓ 20	slightly below R1	12.68 V	Sat 24 Oct, 19:29, indoors
R4	Panel open-circuit voltage (Voc), sun	2 on 1	V⎓ 200	≈ 20–22.0 V (label Voc 22.0 V)	21.8 V	Sun 25 Oct, 10:54, behind window, overcast/rain
R5	Panel short-circuit current (Isc), sun	2 on 1	A 10 (10 A socket)	≤ 1.20 A (label Isc)	0.04 A	Sun 25 Oct, 10:59, behind window, overcast/rain
R6	Battery voltage while charging	2 on 3	V⎓ 20	rises above R1	12.87 V	Tue 27 Oct, 14:15, charging at ~21 mA
R7	Charging current panel → controller	2 in 9	200m (mA), in series	up to ≈ 1 A	20.8 mA	Tue 27 Oct, 14:10, behind window

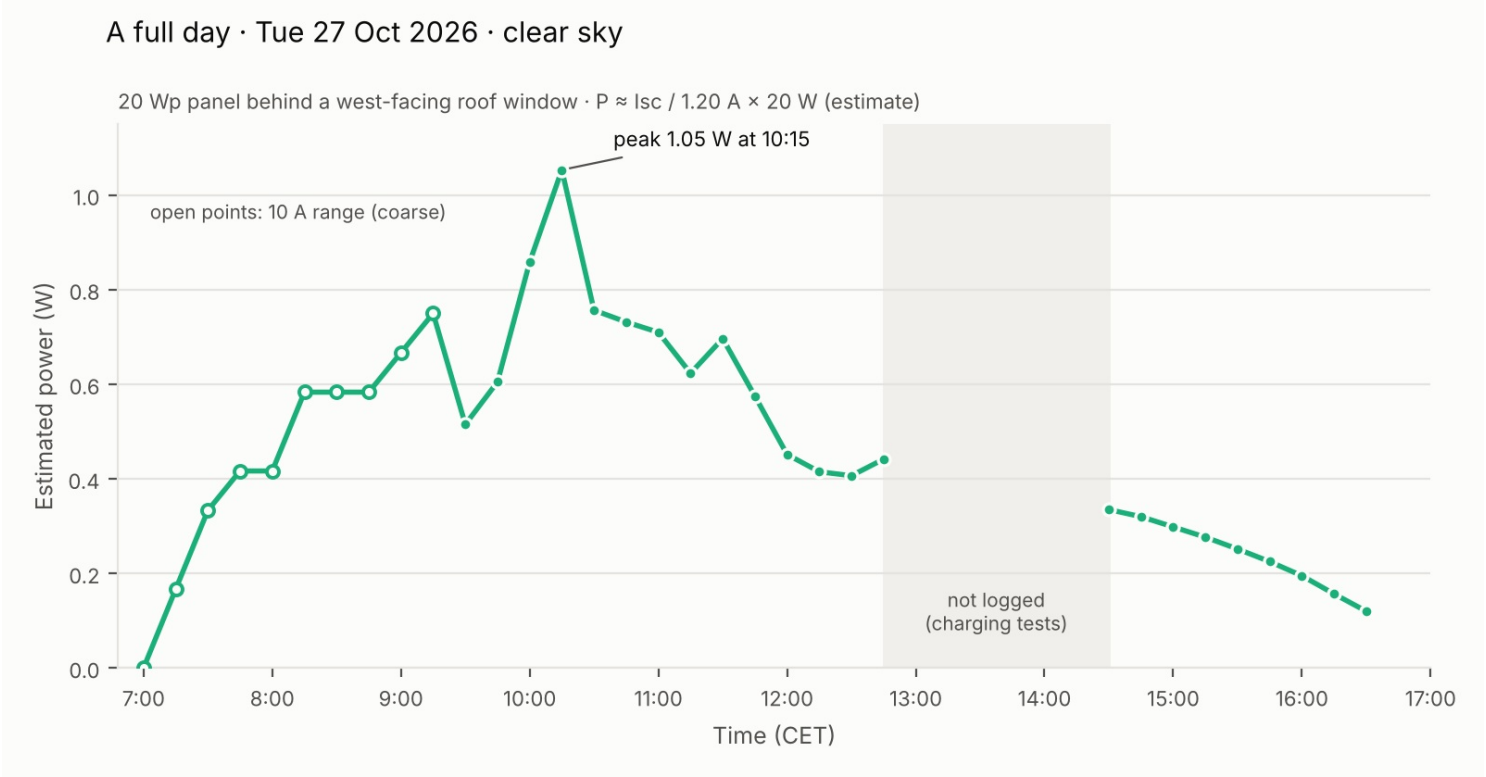
Parts: 1 Solar panel 20 W · 2 Multimeter · 3 Battery 12 V 7.2 Ah AGM · 4 Charge controller PWM 10 A · 5 USB fan · 6 Fuse 5 A · 7 Lever connector · 8 Battery cable · 9 Panel cable



Daily production log

Short-circuit current (Isc) logged by camera every 15 min · panel fixed behind a west-facing roof window (284°) · P ≈ Isc / 1.20 A × 20 W (estimate)





Day	Weather	Logged	Peak	Energy, logged span	Energy, 14:30–16:30
Sun 25 Oct	Overcast, rain (high 10 °C)	11:15–16:30	50 mA at 12:45 (≈ 0.83 W, ≈ 42 W/m²)	≈ 2.0 Wh	0.32 Wh
Mon 26 Oct	Overcast, dry (high 12 °C)	14:30–17:00	62 mA at 14:30 (≈ 1.03 W, ≈ 52 W/m²)	≈ 1.2 Wh	**1.17 Wh (≈ 3.6× Sunday)**
Tue 27 Oct	Clear sky	07:00–16:30 (gap 12:45–14:30)	63 mA at 10:15 (≈ 1.05 W, ≈ 53 W/m²)	≈ 4.4 Wh	0.49 Wh

Weather: Apple Weather, Passau, recorded conditions (Sun, Mon); Tue from photos of the sky. Same panel, same position, same hours: only the weather changes.